Using LLMs for Classification of Text Messages from Experiments

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The state of prompt engineering

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Confusing for authors; confusing for referees.

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- Frontier models (GPT 5.4): robust to formatting, wording, persona, reasoning.
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Tasks	*Promise* (public-goods chat), *Level-k I* (voting), *Level-k II* (coordination) — ∼2,000 items.

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Axes	*content, format, framing, examples, reasoning, information fidelity.*

What we find

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Open matches closed	Qwen 397B = GPT 5.4 with the full codebook.

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On a capable model, with a well-specified prompt:

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Surface choices don't improve on a well-specified codebook.

Smaller / open models remain sensitive to these surface choices.

Prompt = codebook

	P	L_1	L_2
GPT 5.4	**91.2**	**84.0**	**81.7**
GPT 5.4-nano	79.4	78.3	70.7
Qwen 397B	**90.0**	**83.6**	77.8
Qwen 9B	84.1	**82.8**	68.2

Verbatim codebook + its worked examples, no reasoning.

How we reconcile the literature

	Underspecified prompt	*Well-specified prompt*
Small / open model	**very sensitive**	sensitive
Capable frontier model	sensitive	**robust**

Sensitivity comes from either side. Robustness needs both.

Practical implications

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Pick a capable model.

Skip the rest. Formatting, role persona, wording, reasoning.

Results in detail

↗ Content	accuracy surface · variance decomposition
↗ Format	significance table · format scatter
↗ Framing	significance table · per-task scatter
↗ Examples	content vs. examples substitutability
↗ Example formatting	significance table · shot scatter
↗ Reasoning	content vs. reasoning · GPT 5.4 dosage

Thank you.

can-celebi.github.io/prompting

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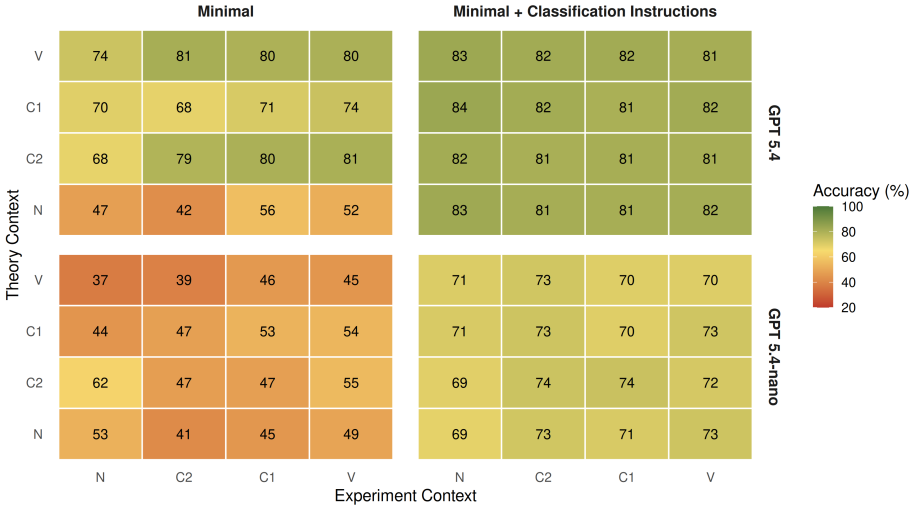
Content · information surface (L_1, GPT 5.4)



↗ variance decomposition ⟷ back

Content · what drives accuracy

	P	L_1	L_2
Classification instructions	2.6 %	**58.3 %**	27.4 %
Theory context	—	8.5 %	9.6 %
Experiment context	3.1 %	1.3 %	1.9 %
Model choice	**94.3 %**	31.9 %	**61.1 %**

On L: classification definitions dominate. On P: model choice does.

↩ back

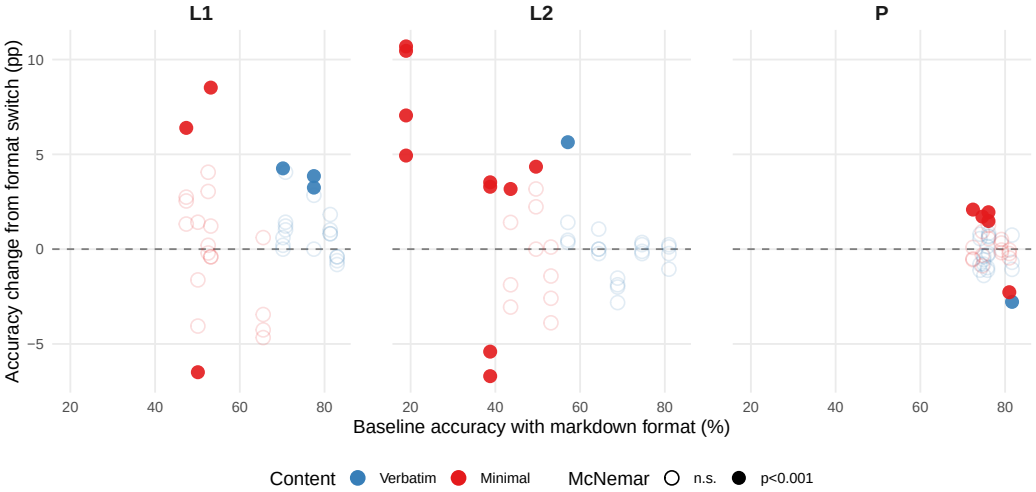
Format · proportion of significant comparisons

	Minimal		Verbatim	
	+	−	+	−
GPT 5.4	**.17**	.08	.00	.08
GPT 5.4-nano	.00	.00	.08	.00
Qwen 397B	.08	.08	.00	.00
Qwen 9B	**.25**	**.17**	.00	.00

Verbatim: nothing significant. Minimal: sparse, inconsistent direction.

↗ scatter · format example ⟷ back

Format · per-task scatter



↩ back

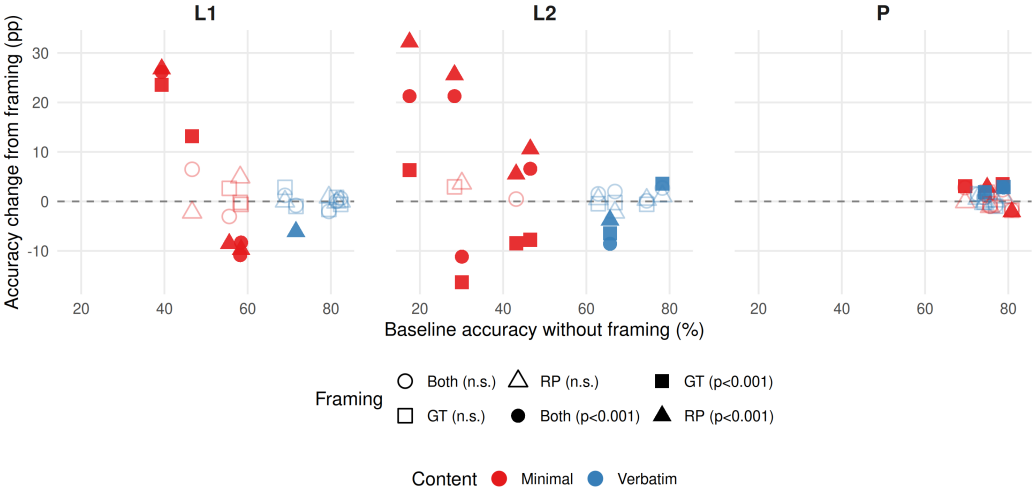
Framing · proportion of significant comparisons

	Minimal		Verbatim	
	+	−	+	−
GPT 5.4	**.33**	**.11**	**.33**	.00
GPT 5.4-nano	**.33**	**.22**	.00	.00
Qwen 397B	**.11**	**.33**	.00	.00
Qwen 9B	**.78**	.00	.00	**.11**

Verbatim: almost nothing significant. Minimal: sparse, inconsistent direction.

↗ scatter ↩ back

Framing · per-task scatter



↩ back

Examples · content and examples substitute

GPT 5.4; similar pattern in the other three models.

	P	L_1	L_2
Δ_{Ex}	$+11.7^*$	$+37.4^*$	$+20.4^*$
$\Delta_{\text{Ex}} - \Delta_{\text{Content}}$	$+11.1^*$	$+3.4$	-10.9^*
$\Delta_{\text{Ex} \mid \text{Content}}$	$+9.9^*$	$+2.6$	$+0.7$
$\Delta_{\text{Content} \mid \text{Ex}}$	-1.2	-0.8	$+11.6^*$

Δ_{Ex}: examples added to minimal. $\Delta_{\text{Ex} \mid \text{Content}}$: examples given verbatim content.

Examples and content carry overlapping information.

↩ back

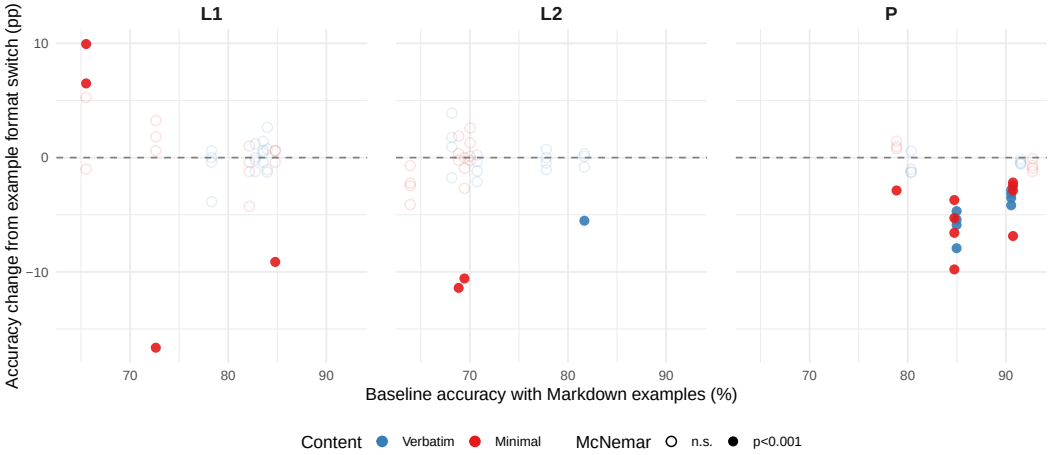
Example formatting · significance

	In-prompt formats				*Multi-turn*			
	Minimal		Verbatim		Minimal		Verbatim	
	+	−	+	−	+	−	+	−
GPT 5.4	.00	.00	.00	.00	.00	**.33**	.00	**.33**
GPT 5.4-nano	.00	.00	.00	.00	.00	**.67**	.00	.00
Qwen 397B	.00	**.33**	.00	**.33**	.00	**.67**	.00	**.33**
Qwen 9B	**.22**	**.33**	.00	**.33**	.00	**.67**	.00	**.33**

In-prompt format: indifferent. Multi-turn (examples outside the system message): hurts.

↗ scatter ↩ back

Example formatting · per-task scatter



↩ back

Reasoning · can it substitute for content?

GPT 5.4.

	P	L_1	L_2
Δ_{Rea}	$+6.6^*$	-11.5^*	$+19.7^*$
$\Delta_{\text{Rea}} - \Delta_{\text{Content}}$	$+5.9^*$	-45.4^*	-11.6^*
$\Delta_{\text{Content} \mid \text{Rea}}$	-6.8^*	$+46.7^*$	$+11.6^*$
$\Delta_{\text{Rea} \mid \text{Content}}$	-0.8	$+1.2$	$+0.0$

Reasoning can't substitute for content; adds nothing once content is present.

↗ effect by reasoning level ↩ back

Reasoning dosage · GPT 5.4 (verbatim content)

	0-shot			*n*-shot		
Step	P	L_1	L_2	P	L_1	L_2
off → low	$+4.6^*$	$+1.6$	-1.9	$+0.8$	$+0.6$	-2.0
low → med	-0.1	$+1.0$	$+2.4$	$+0.5$	$+2.0$	$+1.9$
med → high	-5.7^*	-1.4	-0.5	$+0.0$	-0.6	$+2.0$
high → xhigh	-5.6^*	$+0.0$	$+0.7$	-0.3	-1.4	-0.7

pp, each step relative to the previous level. * = *McNemar* $p < 0.001$.

No incremental step from low to extra-high produces a significant gain.

↩ back